

Using knowledge graph embeddings in data classification and clustering

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Abstract— The knowledge graph is a network of real-world entities and the relationships between them. Graph embedding techniques are used to improve computations in the knowledge graph. The embedding map the knowledge graph into a continuous vector space with low dimensions so that this vector preserves the structural and semantic properties of the graph. The knowledge graph is used in tasks such as link prediction, entity clustering, entity disambiguation, and other downstream tasks. In this paper, knowledge graph embeddings are used in classification and clustering tasks and embeddings are considered as features with which classification models are trained. ComplEx model is used to create knowledge graph embeddings. Also, the problem of fine-tuning hyperparameters of the complEx model is mapped to an optimization problem and the problem is solved with a genetic algorithm.

Keywords—Hyperparameter fine-tuning; Clustering; Classification; Emmbdding; Knowladge graph;

I. INTRODUCTION

The knowledge graph is a network of real-world entities and the relationships between them, created based on knowledge sources such as ConceptNet [1], WordNet [2], Google's Knowledge Vault [3], and so on. Knowledge graph information is stored in a graph database. Information is in standard format RDF¹ and is stored in the form of <subject, predicate, object> triads. To improve computations, graph information is stored in embedded formats. Computers can easily process data that have a numeric format, so graph information is stored in the form of low-level vectors called embedding. Knowledge graph embeddings are may be used for link prediction and knowledge discovery [4]. They can also be used in tasks such as clustering and classification. There are different models for creating knowledge embeddings. Some of these models are TransE [5], ComplEx [6], DistMult [7], HolE [8] which are implemented by the AmpliGraph². In this work, the model ComplEx is used.

In clustering, data are divided into groups or clusters of samples that have similar characteristics. Clustering algorithms are considered unsupervised learning, and the most famous of them is the K-Means clustering algorithm [9], which is used in

the current work. The K-Means clustering algorithm is one of the unsupervised learning algorithms that divides the data set into a certain number of clusters. The main idea in this algorithm is to specify K centers for K clusters.

Classification is a process in which a model is created from labeled samples, and the model can predict the label of samples that do not have a label. In this work, XGBoost algorithm is used to classify the data. It was proposed by the Chen et al [10]. XGBoost provides a parallel tree boosting that Many data science problems are solved with it in a fast and accurate way.

In the current work, the dataset "International Football Results from 1872 to 2019" is used. First, a knowledge graph is created from the dataset. Then, using complEx model, knowledge graph embedding is created. The quality of embeddings is evaluated and finally clustering and classification operations are performed using the created embeddings. The embeddings of each sample of the data set are considered as features of that sample. Figure 1 shows the general steps of the operation performed in this paper.

Also, Genetic algorithm has been used to fine-tune the hyperparameters of the ComplEx model. The basic principles of the genetic algorithm were presented by John Holland et al at the University of Michigan in 1962 to 1965. But its usual form was developed by Goldberg [11]. Genetic algorithm is a meta-heuristic algorithm that is used to optimize and solve problems that do not have a polynomial solution.

The problem of fine-tuning the hyperparameters of the complEx model is mapped to an optimization problem with three variables and genetic algorithm has been used to find the near-optimal values of these hyperparameters. In the mentioned genetic algorithm, the idea of dynamic programming is used. At each step when a solution is evaluated, that solution is stored in a repository along with its evaluation result. When new solutions are generated, the repository is checked and new solutions are sought in the repository. If there are new solutions, there is no need to reevaluate them and call the fitness function. Also, identical solutions cannot be stored in a mating pool, because these solutions cannot create new solutions in the intersection operation. In other words, if two solutions with the same values are combined, a similar solution is created. Therefore, storing similar solutions in the mating pool is prevented.

¹ Resource Description Framework

² <https://docs.ampligraph.org/en/1.4.0/>

AmpliGraph website³ teaches the use of embeddings in data classification and clustering. In the evaluation section, the results obtained in this article are compared with the results of the AmpliGraph website.

The main contributions of this paper are: 1- Mapping the problem of fine-tuning the hyperparameters of the complEx model to an optimization problem with three variables 2- Proposing a genetic algorithm to fine-tune hyperparameters 3- Using the saving-for-the-future idea of dynamic programming to reduce the number of the fitness function calls 4- Using the knowledge graph embedding to train classification and clustering algorithms.

used to predict the result of the matches. Prediction of matches' results is considered and evaluated as a multi-class classification problem.

A. Create knowledge graph

a knowledge graph is created from "International football results" data set⁴. Graph data structure can maintain the relationship between data. In the created knowledge graph, each match is an entity and will be linked to participating teams, characteristics, results and geography. Figure 2 shows the knowledge graph of the 3129 matches. Knowledge graph information is stored based on the RDF standard [12]. RDF is used to store the semantic information of web resources. In this standard, data is stored as a triple <subject, predicate, object>.

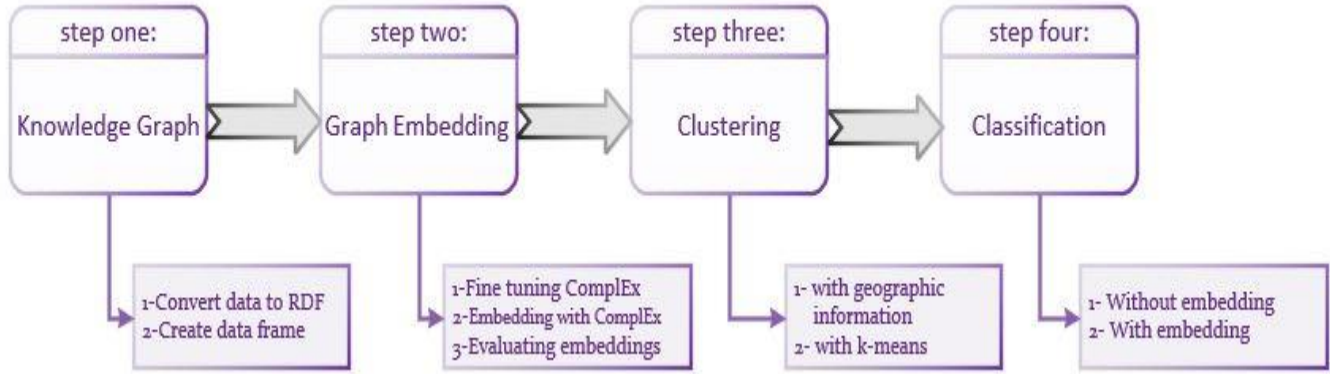


Figure 1: General process of using knowledge graph for clustering and classification of data

I. THE PROPOSED APPROACH

In the proposed approach, as the first step, a knowledge graph is created from the data set. The advantage of the graph over other data structures is that the graph can display the relationship between the data well. In the created knowledge graph, each match is considered as an entity that are connected to match results, geographical information, participating teams and match specifications.

In the second step, the data is divided into training and testing sets and knowledge graph embeddings are trained using the training set.

In the third step, two categories of clusters are considered and the clusters of the first category are compared with the clusters of the second category. To create clusters of the first category, the embeddings are reduced to two dimensions and clustered with k-means algorithm. In the second category of clustering, embeddings are clustered based on continent information. Embedding of teams that are in the same continent are placed in a cluster. Finally, the clusters of the first category (the clusters created by embedding of the knowledge graph) and the clusters of the second category are compared. In the fourth step, knowledge graph embedding is

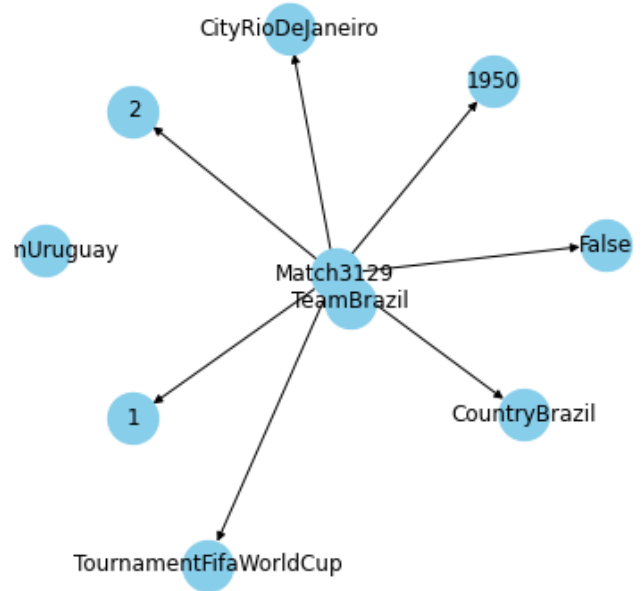


Figure 2: A subgraph of the created knowledge graph from "International football results" dataset

³<https://docs.ampligraph.org/en/1.4.0/tutorials/ClusteringAndClassificationWithEmbeddings.html>

⁴ <https://www.kaggle.com/datasets/martj42/international-football-results-from-1872-to-2017>

Figure 2 shows a subgraph of the created knowledge graph from "International football results" dataset. The subgraph shown in Figure 2 related to match 3129.

B. Embedding knowledge graph

Knowledge graph embedding technique is used to optimize calculations. In this technique, the components of the graph are separately mapped to continuous vectors with low dimensions, such that the structural features and semantic information of the graph are preserved in these vectors.

In the current work, the complEx model [6] is used to create knowledge graph embeddings. In this way, embeddings with complex values are used. Complex embeddings can handle many types of symmetric and asymmetric binary relationships. This approach uses trilinear Hermitian dot product. The time and space complexity of this approach is linear.

II. CLUSTERING OF KNOWLEDGE GRAPH EMBEDDINGS

Knowledge graph embeddings are displayed in a two-dimensional space and clustered with k-means algorithm. Teams are then clustered according to their continent. Finally, the clusters obtained from the k-means algorithm are compared with the clusters obtained through the continents in order to compare the quality of the clusters.

For this, a dictionary is created that contains the embeddings of all teams. In the next step, PCA is used to reduce the dimensions of embeddings. The embeddings are reduced to two dimensions. Then, the embeddings of each team are clustered in six categories using the k-means algorithm. Finally, the clusters created by k-means are compared with the clusters obtained by continents.

III. CLASSIFICATION OF KNOWLEDGE GRAPH EMBEDDINGS

Knowledge graph embeddings can also be used in classification problems. They are used to predict future matches. The classification algorithm used in this part is XGBoost, which is an efficient and flexible algorithm. Embeddings are used as features of the XGBoost algorithm. A three-class problem is considered for prediction of matches. These three classes are: draw, home team loses, home team wins.

Appropriate labels are defined for each sample of the dataset to predict competitions. For this purpose, the number 2 is used to predict a draw, the number 0 is used to predict a team's home win, and the number 1 is used to predict a team's home loss. then, a function is created that extracts the embeddings related to the home and away teams. Finally, the XGBoost algorithm is executed.

The implemented codes in sections II, III and IV are available on GitHub⁵. Colab editor was used for programming. Codes were executed with Google GPU.

IV. COMPLEX FINE-TUNING AS AN OPTIMIZATION PROBLEM

In general, ComplEx model have three hyperparameters that have a great impact on the result: Batches counts (batches_count), number of negative samples (Eta) and dimension of embedding (k). There are two ways to find suitable values for these hyperparameters: 1- Using exact algorithms 2- Using meta-heuristic algorithms. Considering that the mentioned hyperparameters have a wide range of answers and there is no polynomial solution to find suitable values for hyperparameters, metaheuristic algorithms are used. Metaheuristic algorithms provide a good approximation of the optimal solution. To solve the optimization problems with the meta-heuristic algorithm, four steps are performed:

- 1- Create a mechanism to define, store and retrieve the model
- 2- Define a mechanism to create random solutions
- 3- Define the fitness function
- 4- Connect fitness function to model

Fitness function used for guiding meta-heuristic algorithm to best solution is mean reciprocal rank (MRR). The value of MRR is between zero and one. A genetic algorithm is used to fine-tune the hyperparameters of the complEx model. In the next section, this algorithm is briefly explained. Figure 3 presents an overall idea of optimizing decision variables in this work to perform complEx hyperparameter fine-tuning.

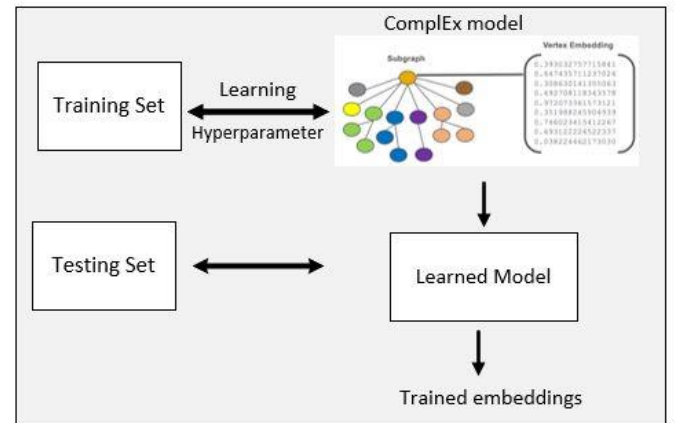


Figure 3: Optimizing decision variables

A. Genetic algorithm

In the genetic algorithm, an initial population of candidate solutions is created and these solutions are evaluated. Next, crossover, mutation and selection operators are executed until the termination condition is satisfied.

Selection operator is an operator that copies the best solutions into a new population. In other words, this operator selects the good chromosomes of the population and places them in a pool called the Mating Pool. The selection method is based on the roulette wheel [13]. In the roulette wheel, the probability of selecting a chromosome to be placed in the mating pool is proportional to the fitness of the chromosome. The probability of selection is calculated from equation 1.

⁵<https://github.com/mohammadRahimi1993/BigData/blob/main/EmbeddingClusteringClassification.ipynb>

$$P_i = \frac{F_i}{\sum_{i=1}^N F_i} \quad (1)$$

In equation 1, N is the population size and P is the probability of selecting a chromosome.

Uniform crossover operator is used to combine two chromosomes [13]. This operation is done with the aim of producing better chromosomes. Combining two chromosomes in the previous generation, causes those new chromosomes be created in the next generation. In a uniform crossover, the old chromosomes are not divided into segments. It is randomly decided that one gene is swapped with the same location on another chromosome.

A uniform mutation operator is applied to the old chromosome to generate a new chromosome with the mutation probability (M_p). The new chromosome is placed in the new population.

Also, In the proposed genetic algorithm, the idea of dynamic programming is used. The solutions evaluated at each step are stored along with their fitness. In the next steps, each solution that is generated is checked before calling the fitness function. If this solution has already been evaluated by the fitness function, the fitness function is not called, otherwise the fitness function is called and the solution is evaluated. All the codes of this section are available on GitHub⁶.

V. EVALUATION

To evaluate the proposed approach, the dataset of "International football results"⁷ has been used, which can be accessed through the Kaggle website.

A. Genetic algorithm experiment results

In this section, the results obtained from the optimization of complEx hyperparameters are presented. The fitness function is defined based on the mean reciprocal rank criterion [14]. This criterion evaluates the solutions produced by the genetic algorithm. If the value of mean reciprocal rank is higher, this criterion is better. figure 1 shows the optimization results.

The results show that the genetic algorithm has been able to fine-tune the hyperparameters of the model well. In the first iteration of the algorithm, the value of the fitness function is 0.139 and in the thirteenth iteration it is 0.179.

B. Evaluation of embedding

To evaluate the quality of embeddings, three criteria, mean reciprocal, mean reciprocal rank and Hit@k have been used [14]. The mean reciprocal (MR) criterion is used to evaluate the ranking components of question-and-answer systems. This criterion is calculated from the equation 2.

$$MR = \frac{1}{|Q|} \sum_{q \in Q} \text{rank}(q) \quad (2)$$

In equation 2, Q is a set of queries, $\text{rank}(q)$ is the rank of each query, and $|Q|$ the size of the test set. During testing, the entities subject and object are predicted.

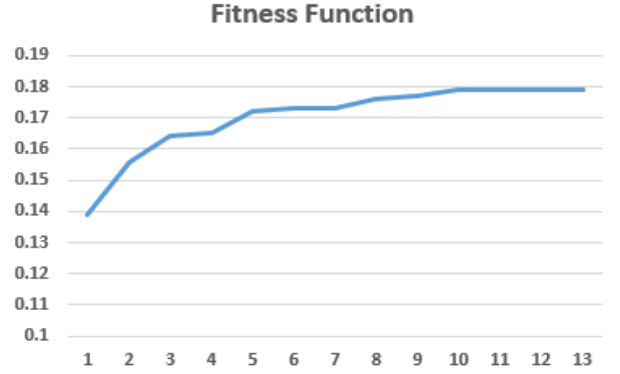


Figure 4: Genetic algorithm experiment result

Another criterion that is used to evaluate embeddings is mean reciprocal rank and is displayed as an abbreviation with MRR and it is calculated from the equation 3.

$$MRR = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}(q)} \quad (3)$$

Hit@k is a probabilistic measure and ranks are considered to be less than or equal to a threshold value. This criterion is calculated from the equation 4:

$$\text{Hit@k} = \frac{|q \in Q: \text{rank}(q) \leq k|}{|Q|} \quad (4)$$

Where k is the number of triplets with the highest score, sorted from largest to smallest. K can be 1,3,5,10. If the value of Hit@k is higher, this criterion is better. Finally, the values of the mentioned criteria according to the test set can be seen in the table 1:

Table 1: Evaluation criteria for embeddings

	MRR	MR	Hits@10	Hits@3	Hits@1
AmpliGraph website	0.26	4365.06	0.36	0.29	0.19
Current Research	0.29	3065.02	0.41	0.34	0.22

The results show that the correct entity is in the top-10 possibilities in 41% of cases. Also, this value is better than the value obtained on the AmpliGraph website. Also, the value of the MRR is 0.29, which is better than the value obtained in the training on the AmpliGraph website. According to the results, it can be concluded that the ComplEx model is well trained and can be used in classification and clustering tasks.

C. Evaluation of clustering and classification

Figures 5 and 6 show the clusters of the first and second categories. As can be seen, the embeddings have been able to

⁶<https://github.com/mohammadRahimi1993/BigData/blob/main/CheckGAFinETune.ipynb>

⁷<https://www.kaggle.com/martj42/international-football-results-from-1872-to-2017>

learn the geographical similarity of each team very well. Note that the information about each team's continent is not explicitly included in the dataset. It can also be seen that K-means has found clusters very similar to the clusters of continents. This shows that in 200 dimensions, similar embeddings are close to each other in terms of distance.

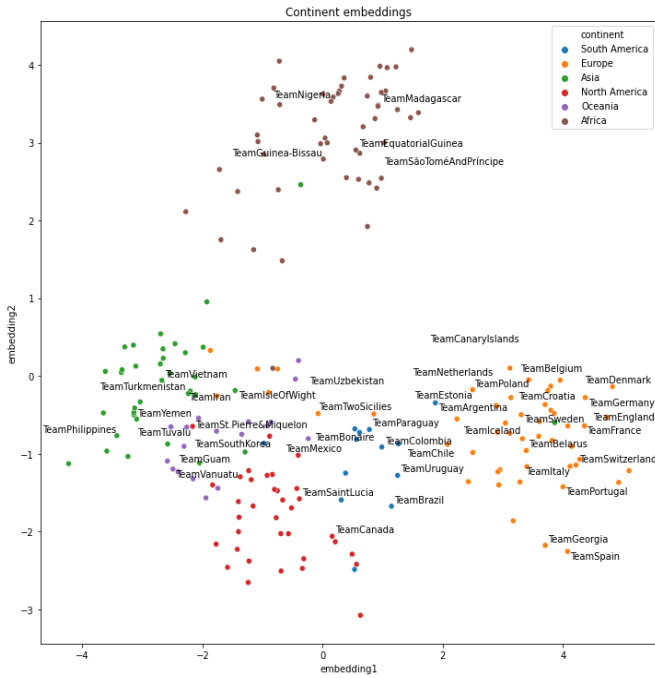


Figure 5: Clustering based on continents information

Also, in order to determine the effect of embeddings in data classification, the accuracy of XGBoost model was measured before and after training by embedding. The baseline accuracy for the mentioned problem is 47%. this accuracy is the frequency of the most frequent class (home team wins). While with knowledge embeddings, this accuracy has increased.

Table 2: Classification accuracy with algorithm XGBoost

	Baseline	Embedding
AmpliGraph website	47.1%	53.7%
Current Research	47.1%	55.1%

Table 2 shows the results of data classification. if the classification accuracy of this article is compared with the training posted on the AmpliGraph website, the classification in this article is about 2% more accurate.

VI. CONCLUSION

In this article, knowledge graph embeddings are used in data classification and clustering. The ComplEx model has been used to create embeddings. The results show that the use of embeddings can improve the accuracy of classification and clustering data.

Another work that has been done is fine-tuning the hyperparameters of the complEx model using a genetic

algorithm. An optimization problem was defined with complEx hyperparameters and this problem was solved using genetic algorithm. The results show that the genetic algorithm has been able to fine-tune the hyperparameters of complEx model well. In the used genetic algorithm, the idea of dynamic programming is used and it is not allowed to store duplicate solutions in the mating pool. Using the idea of dynamic programming makes the genetic algorithm run faster.

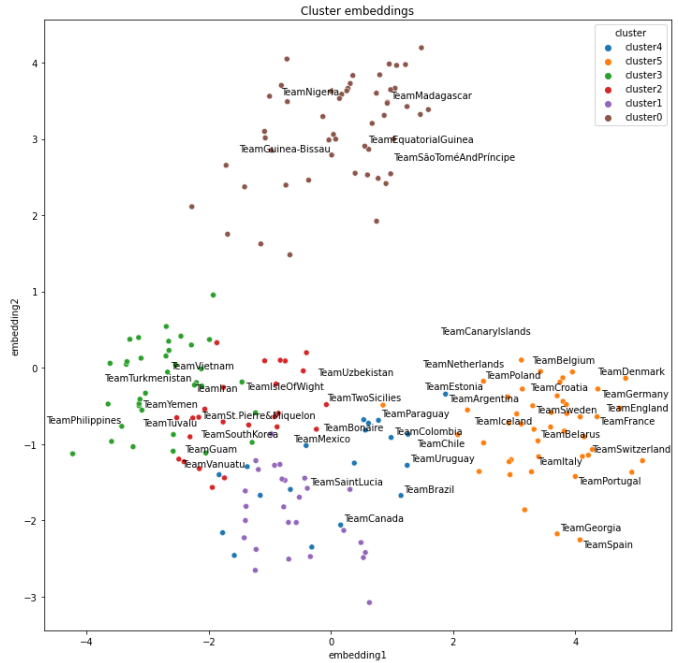


Figure 6: Clustering with k-means algorithm

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